

Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

PhishGuard

{ A Machine Learning-Based Phishing Detection System }

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- ***Phishing*** is one of the most common and damaging cyber threats.
- It uses fake websites and messages to *steal* credentials and ***sensitive data***.
- Traditional blacklist-based defenses are weak against zero-day and adaptive phishing.
- **PhishGuard** addresses this by using *machine learning* for phishing detection.

Existing phishing detection systems have major limitations:

1. They fail on zero-day phishing domains.
2. Heuristic systems generate many false positives.
3. Single-feature detection is not reliable.
4. Real-time detection must remain fast and efficient.

- **Build a machine learning-based phishing detection system**
Detect malicious URLs using intelligent feature analysis instead of static blacklists.
- **Extract multi-dimensional phishing features**
Analyze lexical, structural, and semantic URL characteristics.
- **Improve detection accuracy and reduce false positives**
Use supervised learning and ensemble techniques for reliable classification.
- **Support real-time phishing detection**
Enable low-latency prediction suitable for browser or SOC deployment.
- **Enhance explainability and analyst visibility**
Use SHAP-based interpretation to explain phishing predictions.

Project Structure

src/

- └─ features/ → URL & domain feature extraction
- └─ training/ → ML model training pipeline
- └─ evaluation/ → Metrics & performance analysis
- └─ explainability/ → SHAP explanation module
- └─ api/ → Streamlit dashboard integration
- └─ models/ → Saved trained models
- └─ data/raw/ → Original phishing datasets
- └─ data/processed/ → Feature-engineered datasets
- └─ reports/ → Evaluation graphs & outputs

Architecture Type:

Modular machine learning architecture separating feature extraction, model training, evaluation, and deployment layers.

- **Input Layer**

- Raw URLs, domain information, and webpage content

- **Feature Engineering Layer**

- Lexical feature extraction
- Structural URL analysis
- Semantic phishing indicators

- **Machine Learning Layer**

- Random Forest
- SVM
- ANN
- XGBoost Ensemble

- **Explainability Layer**

- SHAP-based feature contribution analysis

- **Output Layer**

- Phishing / Legitimate classification
- Risk scoring and analyst visibility

Architecture Type:

Layered machine learning architecture separating data processing, feature extraction, model inference, and explainability modules.

TECHNOLOGY STACK

Backend / ML

- Python 3.11
- Scikit-learn
- XGBoost
- SHAP

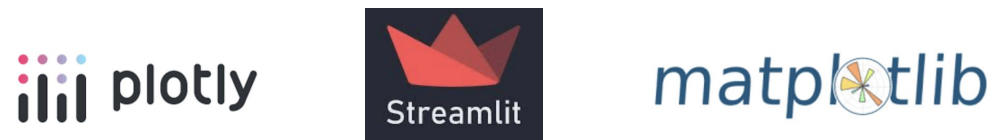


Data Processing

- Pandas
- CSV-based phishing dataset storage
- NumPy
- SciPy

Frontend

- Streamlit
- Plotly
- Matplotlib



Tools

- Joblib
- Visual Studio Code
- Git & GitHub



Feature engineering is the intelligence layer of the system. Consists:

- lexical analysis
- host-based analysis
- HTML/content analysis
- dimensionality reduction

Feature Extraction

- TLDEExtract
- BeautifulSoup4
- Requests
- URL Parsing Utilities

Machine Learning Models

<u>ML-MODELS</u>	<u>PURPOSE</u>
Random Forest	<i>Best balance</i>
SVM	<i>Margin-based classification</i>
ANN	<i>High recall</i>
XGBoost	<i>Gradient boosting</i>

Storage Organization

data/

|— raw/

| |— phishing_dataset.csv

|— processed/

| |— features.csv

models/

|— phishguard_rf.joblib

|— scaler.joblib

|— feature_names.joblib

/reports/

|— figures/

|— confusion_matrix.png

|— roc_curve.png

|— shap_analysis.png

src/

|— features/

|— training/

|— evaluation/ |— explainability/ |— api/

- Multi-feature phishing URL detection
- Lexical, structural, and semantic analysis
- Ensemble machine learning classification
- Real-time phishing prediction
- SHAP-based explainability
- Suspicious TLD and IP detection
- Brand impersonation detection
- Feature-engineered URL intelligence
- Low-latency detection pipeline
- Streamlit-based analyst dashboard



Performance Metrics

- **Accuracy**

- Measures overall phishing detection performance

- **Precision**

- Evaluates correctness of phishing predictions

- **Recall**

- Measures ability to detect malicious URLs

- **F1-Score**

- Balances precision and recall performance

- **ROC-AUC**

- Evaluates classifier discrimination capability

- **Confusion Matrix**

- Visualizes phishing vs legitimate classifications

- **Low-Latency Processing**

- Fast feature extraction and prediction pipeline

- **Model Robustness**

- Improved detection against unseen phishing patterns

Results & Experimental Analysis



PhishGuard

ML-powered phishing detection with SHAP explainability

About

PhishGuard analyses URLs using 27 hand-crafted features across three categories:

- **Lexical** — URL string patterns
- **Structural** — domain and TLD signals
- **Semantic** — brand impersonation & keywords

Model

Model loaded

Type: RandomForestClassifier

Deploy

https://mail.google.com/mail/u/0/#inbox/FMfcgZvPBHhXrWTzbSDsndsGW

Analyse

✓ Legitimate

Phishing probability: 15.3%

15.3%

RISK LEVEL

SAFE

PHISH PROB

15.3%

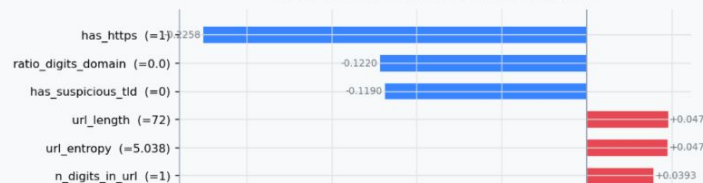
LEGIT PROB

84.7%

FEATURES USED

27

Feature Contributions to This Prediction



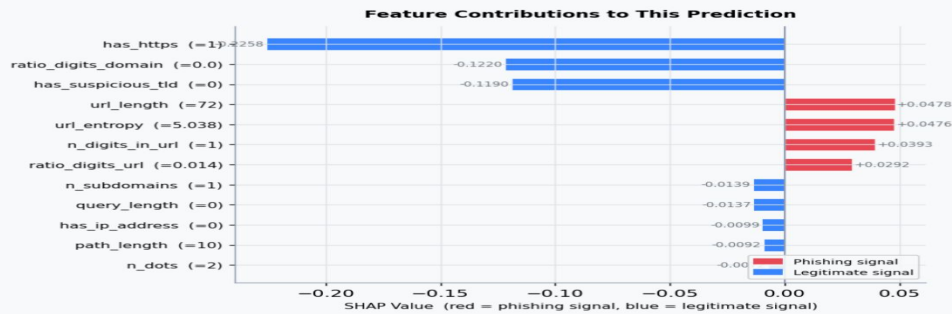
has https

HTTPS present (encrypted / trusted connection)
value=1 | SHAP -0.2258

ratio digits domain

High digit ratio in domain name

Results & Experimental Analysis



has https

HTTPS present (encrypted / trusted connection)
value=1 | SHAP -0.2258

ratio digits domain

High digit ratio in domain name
value=0.0 | SHAP -0.1220

has suspicious tld

Domain uses a high-risk TLD (.xyz .tk .ml etc.)
value=0 | SHAP -0.1190

url length

URL is unusually long
value=72 | SHAP +0.0478

url entropy

High randomness in URL (auto-generated domain)
value=5.0376 | SHAP +0.0476

n digits in url

Many digit characters in URL
value=1 | SHAP +0.0393

ratio digits url

High digit ratio in URL

VERDICT: Legitimate | Risk: SAFE | Phishing probability: 15.3%

URL (truncated): <https://mail.google.com/mail/u/0/#inbox/FMfcgzQgLRssZvPBHhXrWTzbSDsnds>

Top signals driving this decision:

[HIGH] Connection uses HTTPS (encrypted / trusted)
(feature=has_https, value=1, SHAP=-0.2258 -> LEGITIMATE)
[HIGH] Domain name has a high proportion of digits
(feature=ratio_digits_domain, value=0.0, SHAP=-0.1220 -> LEGITIMATE)
[HIGH] Domain uses a high-risk TLD (.xyz, .tk, .ml, etc.)
(feature=has_suspicious_tld, value=0, SHAP=-0.1190 -> LEGITIMATE)
[MEDIUM] URL is unusually long
(feature=url_length, value=72, SHAP=+0.0478 -> PHISHING)
[MEDIUM] URL string has high randomness (suggests auto-generated domain)
(feature=url_entropy, value=5.0376, SHAP=+0.0476 -> PHISHING)

Base rate (expected output without features): 0.5002

NOTE: This tool is for defensive triage only. Always confirm before blocking.

Evaluation metrics screenshot

```
=====
PhishGuard -- Full Evaluation Suite
=====
Model      : RandomForestClassifier
Test samples : 1000
Features   : 27

=====
Classification Report
=====
```

	precision	recall	f1-score	support
Legitimate	1.0000	1.0000	1.0000	500
Phishing	1.0000	1.0000	1.0000	500
accuracy			1.0000	1000
macro avg	1.0000	1.0000	1.0000	1000
weighted avg	1.0000	1.0000	1.0000	1000

```
ROC-AUC score      : 1.0000
Average Precision (AP) : 1.0000

=====
Summary Metrics
=====
```

Accuracy	1.0000	
Precision	1.0000	
Recall (TPR)	1.0000	
F1-Score	1.0000	
Specificity (TNR)	1.0000	
False Positive Rate	0.0000	
False Negative Rate	0.0000	
ROC-AUC	1.0000	
Avg Precision (AP)	1.0000	
MCC	1.0000	

```
Generating plots ...
Saved: /Users/vansh/Projects/phishguard/reports/figures/confusion_matrix.png
Saved: /Users/vansh/Projects/phishguard/reports/figures/roc_curve.png
Saved: /Users/vansh/Projects/phishguard/reports/figures/precision_recall_curve.png
Saved: /Users/vansh/Projects/phishguard/reports/figures/feature_importance.png

All evaluation artefacts saved to: /Users/vansh/Projects/phishguard/reports/figures
```

- **Browser-side phishing detection**
 - Real-time URL scanning before page access
- **Email Gateway Integration**
 - Detects phishing links inside incoming emails
- **SIEM / SOC Integration**
 - Sends suspicious URLs for analyst triage
- **Threat Intelligence Support**
 - Assists security teams with phishing analysis
- **Explainable Security Alerts**
 - SHAP-based reasoning for each prediction
- **Low-Latency Inference**
 - Suitable for real-time defensive deployment
- **Scalable Defensive Architecture**
 - Can integrate with enterprise monitoring systems

Explainability & SHAP

- **SHAP-based explainability framework**
 - Explains phishing predictions using feature importance
- **Feature Contribution Analysis**
 - Shows which URL features influenced classification
- **Analyst-Friendly Interpretation**
 - Helps SOC analysts understand detection reasoning
- **Transparent Machine Learning Decisions**
 - Reduces black-box behavior in cybersecurity systems
- **Visual Risk Analysis**
 - Highlights suspicious phishing indicators visually
- **Improved Trust and Reliability**
 - Supports explainable and accountable AI security models
- **Faster Incident Investigation**
 - Assists analysts during phishing threat analysis

- Handling dataset imbalance
- Designing effective phishing features
- Reducing false positives
- Managing adversarial phishing patterns
- Extracting reliable URL intelligence
- Maintaining low-latency prediction
- Feature engineering and preprocessing complexity
- Model tuning and hyperparameter optimization
- Integrating explainability with ML predictions
- Building a scalable defensive architecture

- Browser extension integration for live phishing detection
- Continual learning for adaptive threat evolution
- Real-time cloud-based deployment architecture
- Detection of SMS and social-media phishing attacks
- Integration with SIEM and enterprise SOC platforms
- Advanced deep learning-based phishing analysis
- Improved explainable AI visualization techniques
- Threat intelligence feed integration
- Mobile device phishing protection
- Automated phishing incident response workflows

- PhishGuard is a machine learning-based phishing detection system designed for modern cyber threats.
- The system combines lexical, structural, and semantic feature analysis for accurate phishing classification.
- Ensemble learning improves robustness, detection accuracy, and reliability against unseen phishing attacks.
- SHAP-based explainability enhances analyst visibility and trust in model predictions.
- The Project demonstrates a practical, scalable, and real-time defensive cybersecurity solution suitable for SOC deployment.

Thank You